

AUTOSAR High-Level Design (HLD) Specification

Domain: Powertrain & Chassis Control Domain | Release 2.0 (ADAS & Defect Remediation)

Document ID: DOC-AUTOSAR-PT-V2	Target ECU: TriCore TC397x (Domain Controller)
Standard: AUTOSAR Classic Platform 4.4.0	Classification: ASIL-D / ISO 26262
Author: Powertrain Architecture Guild	Release Date: 2026-06-20 (Revision 2.0)

1. System Architecture Overview & Revision Summary

Revision 2.0 introduces ADAS Level-2 integration by incorporating LaneKeepAssistSWC, resolves the undefined provider dependency for throttle signals via PedalInterfaceSWC, formally declares SteeringAngleSensorSWC, and corrects the data-type mismatch on the battery current interface between BatteryManagementSWC and TransmissionControlSWC.

2. Software Component: BrakeControlSWC

Component Type: Application Software Component (ApplicationSWC)

Execution Periodicity: 10ms | **Safety Level:** ASIL-D

Description: Responsible for friction braking, regenerative braking demand, and ADAS emergency deceleration blending.

Port Name	Port Type	Interface Name	Signal / Element	Data Type	Target / Provider
RPort_VehicleSpeed	R-Port (Require)	If_VehicleSpeed	VehicleSpeed_kph	float32	WheelSpeedSensorSWC
PPort_BrakeTorqueRequest	P-Port (Provide)	If_BrakeTorque	BrakeTorque_Nm	uint16	EngineManagerSWC
RPort_SteeringAngle	R-Port (Require)	If_SteeringAngle	SteeringAngle_deg	float32	SteeringAngleSensorSWC
RPort_ADASEmergencyBrake	R-Port (Require)	If_ADASDecelRequest	DecelRequest_mps2	float32	LaneKeepAssistSWC

3. Software Component: SteeringAngleSensorSWC (NEW)

Component Type: Sensor/Actuator Software Component (SensorActuatorSWC)

Execution Periodicity: 10ms | **Safety Level:** ASIL-D

Description: Calibrates and transmits real-time steering wheel angular position and rotational velocity.

Port Name	Port Type	Interface Name	Signal / Element	Data Type	Target / Provider
PPort_SteeringAngle	P-Port (Provide)	If_SteeringAngle	SteeringAngle_deg	float32	BrakeControlSWC, LaneKeepAssistSWC

4. Software Component: PedalInterfaceSWC (NEW)

Component Type: Sensor/Actuator Software Component (SensorActuatorSWC)

Execution Periodicity: 5ms | **Safety Level:** ASIL-B

Description: Acquires dual-channel accelerator pedal potentiometers and broadcasts plausibility-checked throttle demand.

Port Name	Port Type	Interface Name	Signal / Element	Data Type	Target / Provider
PPort_ThrottleDemand	P-Port (Provide)	If_ThrottleDemand	ThrottlePercent	uint8	EngineManagerSWC

5. Software Component: LaneKeepAssistSWC (NEW)

Component Type: Application Software Component (ApplicationSWC)

Execution Periodicity: 20ms | **Safety Level:** ASIL-D

Description: Processes front-facing camera lane boundaries, monitors driver steering torque, and requests autonomous corrective braking or steering torque.

Port Name	Port Type	Interface Name	Signal / Element	Data Type	Target / Provider
RPort_SteeringAngle	R-Port (Require)	If_SteeringAngle	SteeringAngle_deg	float32	SteeringAngleSensorSWC
PPort_ADASDecelRequest	P-Port (Provide)	If_ADASDecelRequest	DecelRequest_mps2	float32	BrakeControlSWC

6. Software Component: TransmissionControlSWC (UPDATED)

Component Type: Application Software Component (ApplicationSWC)

Execution Periodicity: 20ms | **Safety Level:** ASIL-C

Description: Corrected RPort_BatteryCurrent data type from float32 to uint16 matching BatteryManagementSWC.

Port Name	Port Type	Interface Name	Signal / Element	Data Type	Target / Provider
RPort_EngineSpeed	R-Port (Require)	If_EngineSpeed	EngineSpeed_rpm	uint16	EngineManagerSWC
RPort_BatteryCurrent	R-Port (Require)	If_BatteryCurrent	Current_A	uint16	BatteryManagementSWC
PPort_GearPosition	P-Port (Provide)	If_GearStatus	CurrentGear	uint8	InstrumentClusterSWC